Python in Action-Exploring Object-Oriented Programming in the Real World

Object Oriented Programming Exercise 1

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Abstract

Object Oriented Programming (OOP) is a way of writing programs where we think of things as objects. Instead of just writing many functions, we group data and functions together as objects. This paper is a survey paper which means it reviews what other people have written about how Python uses OOP in real life. I looked at different articles, books and documentation to see where Python OOP is used. The findings show that Python OOP is used in web development like Django, in data science like Pandas, in AI like PyTorch and also in business systems like Odoo which is used in Uganda. Python is good for OOP because it is easy to read and easy to learn compared to Java or C++.

1. Introduction

Object Oriented Programming was introduced because programmers had problems managing big programs. When programs become big, it becomes hard to maintain them, use them or even make sense of them if everything is just functions. So, OOP was introduced to make programs look like real life things.[1]

In OOP we have things called classes and objects. A class is like a blueprint or a form, and an object is the real thing made from that blueprint. For example, if Class is "Student", then Martha is an object of that class. Python was created by Guido van Rossum in 1991, and it focuses on making code easy to read. In Python everything is an object which makes OOP natural. A survey paper does not do new experiments, but it summarizes what others have already done. Therefore, in this paper I am summarizing how Python OOP is used in real world.[2][3]

2. Body of the Paper

2.1 Main Concepts of OOP in Python

According to different books, OOP has five main pillars:[1][4]

a) **Class and Object**:
A class is a blueprint. For example, on GitHub, a repository template can be seen as a class and my repository Martha-Valerie-Namatovu/OOP is an object. In Python we create a class using the keyword "class" which is simpler than in Java.[4]

b) **Encapsulation:**

This means hiding private data. Like how we hide our passwords on social media, emails and the like so that others cannot see them. In Python we use _ or __ to show that something is private. This is important in banking systems.[5]

c) **Inheritance**:
This is when a child class gets things from a parent class. It helps us not to repeat code. For example, a UCU Student can inherit from Student and we add a registration or student number. Django uses inheritance a lot to reduce code.[6]

d) **Polymorphism:**
This means same name but different behaviour. Like the word "login" can behave differently for two Instagram accounts. In Python this is done through duck typing.[4]

e) **Abstraction:**
This means hiding complicated details and showing only important parts. Python has Abstract Base Classes for this.[5]

2.2 How Python OOP is Used in Real Life

a) **Web Development:**
Frameworks like Django and Flask are made using OOP. In Django, models, views and forms are all classes. Instagram was built with Django and it shows OOP can handle billions of users.

b) **Data Science and AI:**
Libraries like Pandas have a DataFrame object and Scikit-learn has model objects with methods like fit() and predict(). TensorFlow and PyTorch also use classes to make neural networks.

c) **Business Systems**:
In Uganda many businesses use Odoo ERP which is written in Python. It models real things like customer, invoice, employee as classes. Also small shops can use Python OOP to manage products.

d) **Games and Apps**:
Kivy and Pygame use OOP where every button or character is an object.[6][7][8][4]

2.3 Advantages and Disadvantages

Advantages

a) The code can be reused.
b) It is easy to maintain and teams can work together.

c) Python code is easy to read.

Disadvantages

a) Python does not have real private variables, it only uses convention, and multiple inheritance can be confusing sometimes.

b) Python is slower than C++.[1][5]

3. Conclusion

In conclusion, this survey shows that Python has made OOP easy and practical. It takes the theory of OOP and uses it to build things we use every day like games, business systems, social media applications and other things. From my own example, my GitHub repository "Object Oriented Programming" is an object that I created from GitHub class template. So, OOP is not just theory, it is how we model real life in code. In future, more research can be done on how Python OOP will work with AI tools in Africa.

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